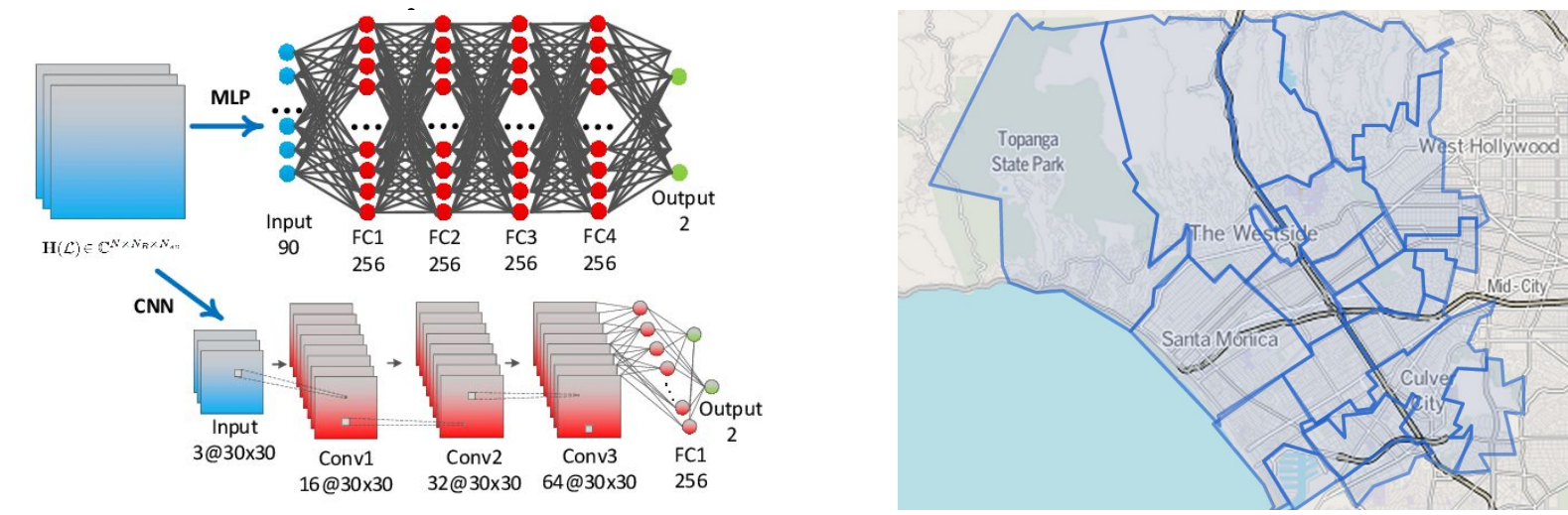


InfernoCommand: A Fire-Relative Reinforcement Learning Framework for Multi-Resource Wildfire Response on a Real Los Angeles Terrain Model

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Background

- Wildfires** are getting more **frequent** and **destructive**. Incident commanders must send **limited** resources, like **ground crews** and **helicopters**, to fight a fire that keeps spreading and changing direction, under **extreme time pressure**.
- Bad allocation** decisions cost structures and lives, and a human commander can only track **so many** moving pieces at once.
- Our course required combining **four core** ML ideas into one agent: **CNN, MLP, Classification**, and **Reinforcement Learning**.
- InfernoCommand** trains this agent on a real 3D model of **LA's Westside** (Topanga, Brentwood, Bel-Air, Westwood, UCLA), built from **real terrain**, road, building, and population data.
- The project is grounded in the **real Palisades Fire** (see Case Study), so the agent is tested against conditions from an actual, historically **destructive** fire.
- The main goal is **generalization**: a policy that only works where it was trained isn't useful in real life, since fires never start in the



Existing Research

- Deep Q-Networks** and **actor-critic** methods have shown that agents can learn to make sequential decisions by **interacting** with an **environment** and getting **rewarded** for good outcomes.
- Prior disaster-response** and **grid-based dispatch** models typically use a fixed set of zones or grid cells as the action space, letting the agent **pick** which zone to send a resource to.
- Fire-spread models** in the literature are usually driven by wind, slope, and fuel type, with real fires known to jump firebreaks through wind-blown embers, a well-documented behavior in fires like the **Palisades Fire**.
- Most **existing** wildfire RL work is tested on a single training scenario or a simplified grid, without much **focus** on whether the learned policy holds up on a fire that starts **somewhere new**.

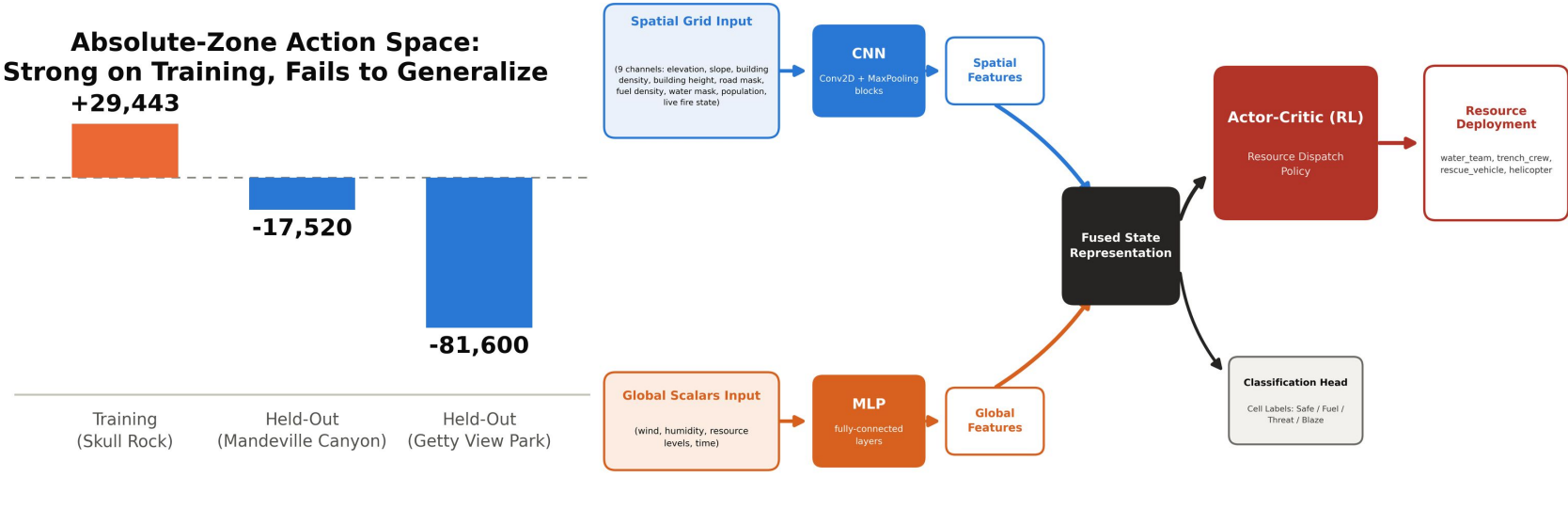
Current Prediction Approaches

System	Approach	Reported Performance
SimFire / SimHarness (MITRE, 2023)	RL land-management agent	Not Reported-Open Source
Cell2Fire + DRL Firebreak Placement (2025)	RL firebreak placement	1.6-1.7% better burned area vs. heuristic, 4.8-6.8% better vs. random
FireCastRL (2026)	Ignition forecast + RL helitack suppression	Reports time-to-containment as a metric type, but no headline % published in available text

Limitation: Existing systems operate over a fixed spatial index (grid cell or zone) and mostly report framework capability or narrow improvement margins rather than generalization to unseen ignition points, the exact gap InfernoCommand's fire-relative action space targets.

Research Gap

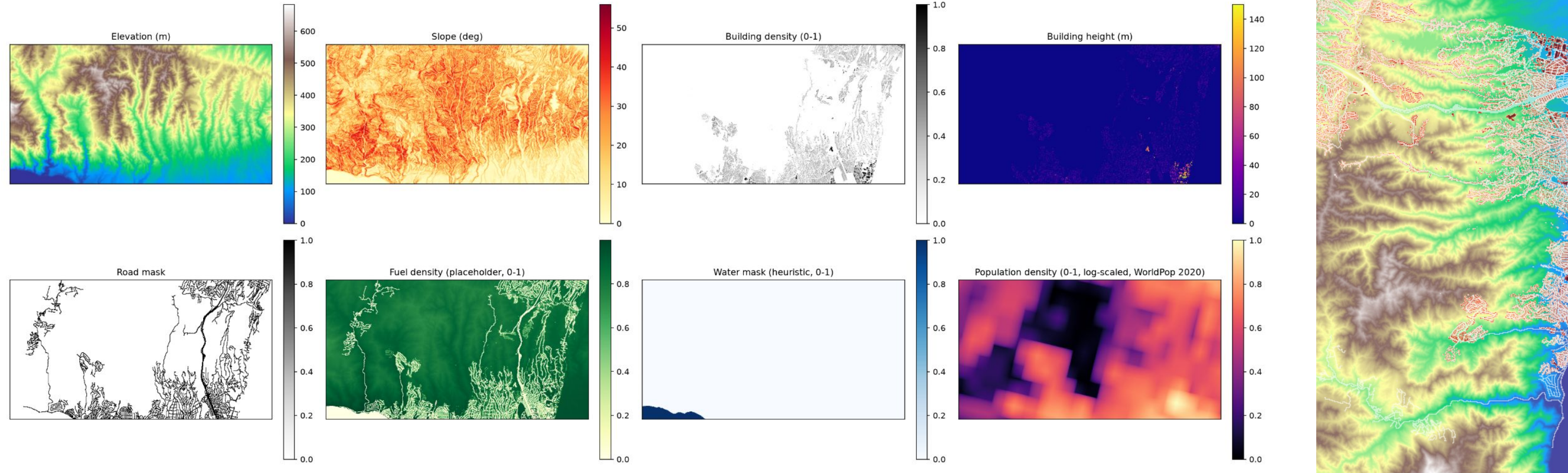
- Absolute zone-index** action spaces are inherently **memorizable**. Telling the agent to "**dispatch to zone 18**" only works because that zone happens to hold the fire in **one** training scenario, so it can't transfer to a fire starting **anywhere else**.
- Real fires** don't start in the same place **twice**. A policy that only works at **one** ignition point has no **real-world value**, no matter how well it scores there.
- Most **existing** wildfire RL work (see Existing Research) reports results on a **single fixed scenario** or **simplified grid**, without rigorously testing whether the learned policy holds up on a held-out, **unseen ignition point**.
- Early attempts** on this project using an absolute zone-based action space confirmed this exact **failure** mode: **strong** performance on the training scenario, but a **severe** drop in reward and containment on **held-out locations**.
- This project directly targets that **gap**: building a dispatch policy that **generalizes** to new ignition points instead of **memorizing** one.



Data Sources & Environment

The **simulation** runs on a real **595×316** grid covering **LA's Westside corridor** (Topanga, Brentwood, Bel-Air, Westwood, UCLA), built entirely from **real data**, not synthetic terrain. Roads, buildings, elevation, weather, and population layers are pulled from **OSMnx**, **LA GeoHub** LARIAC4, USGS 3DEP, NOAA ASOS, and **US Census** sources.

Resource dispatch is grounded in **8** real LAFD stations and **15 real units** (water teams, brush crews, rescue vehicles, helicopters), matched to **actual** station locations and apparatus **types**.

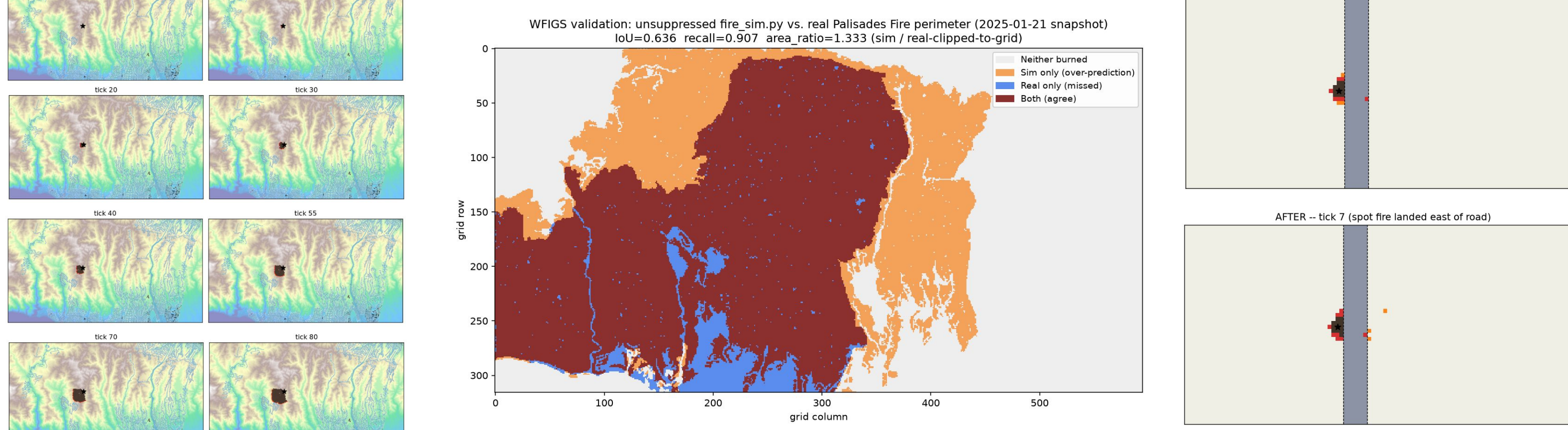


Palisades Case Study & Fire-Spread Validation

The **simulation** is grounded in the **real Palisades Fire** (Jan 7-31, 2025): **~23,400 acres burned**, **~6,800 structures destroyed**, **12 deaths**, driven by **real** Santa Ana conditions — humidity crashed to 0.67%, sustained winds of 25.3 mph.

Fire spreads through a **physics-based rule** combining wind, slope, fuel, and road resistance, including **ember spotting**: burning cells can ignite fuel several cells **downwind**, jumping roads at **high wind speeds**.

Validated against the real historical **Palisades burn perimeter** (WFIGS data): IoU 0.636, **recall 90.7%**, confirming the simulated fire behaves like the **real fire** and is not plausible looking.



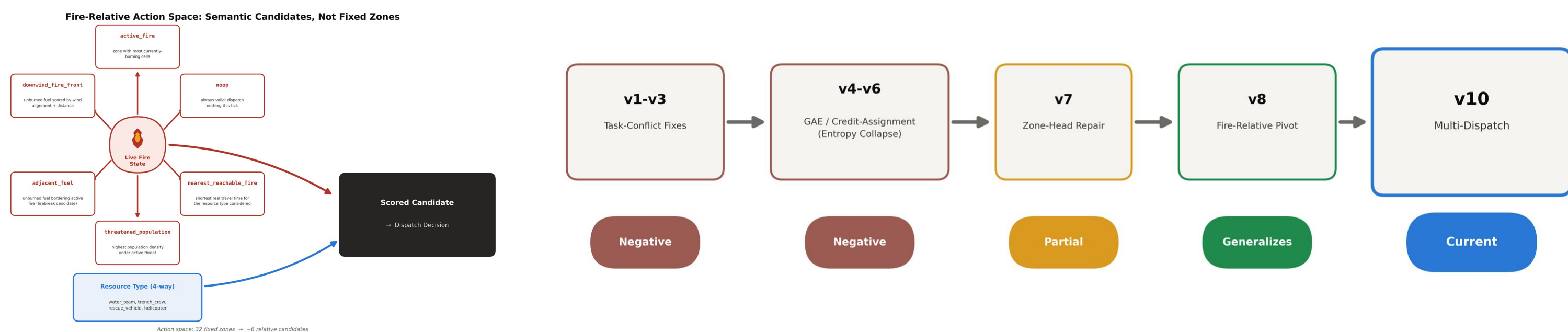
Investigation

The model **fuses** a **CNN** (spatial grid), an **MLP** (weather/resource scalars), a **classification head** (per-cell Safe/Fuel/Threat/Blaze labels), and an **actor-critic** RL policy that **dispatches** resources.

Early attempts using absolute zone indices as the action space **memorized** the training scenario and **collapsed** on held-out fires (see Research Gap).

A deep investigation across **seven model versions** traced a separate **training instability** to a specific cause: a rewrite introducing **GAE** (Generalized Advantage Estimation) made the critic's value estimates **anti-correlated** with reality in short, near-instantly-contained episodes — exactly the trajectories a good policy most needs **reinforced**.

This pointed to a **deeper issue**: the real bottleneck wasn't the **RL algorithm**, it was the **action space itself**.



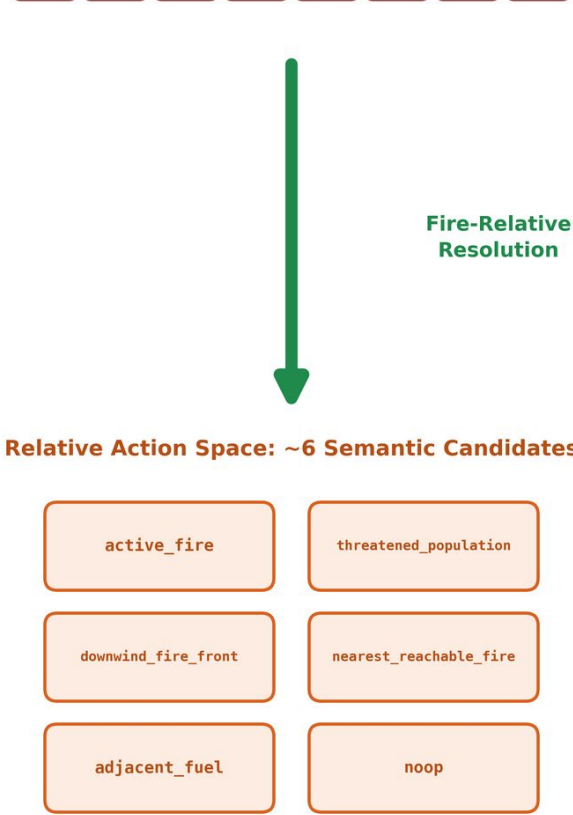
The Relative-Action Model (v8)

Instead of choosing a **fixed zone index**, the model resolves the live fire state into up to **6 semantic** candidates each tick, wherever they currently are: **active_fire**, **downwind_fire**, **front**, **adjacent_fuel**, **threatened_population**, **nearest_reachable_fire**, and **noop**.

The model first picks a **resource type** (water team, brush crew, rescue vehicle, or helicopter), then scores these candidates using each one's own **local features** (fire, fuel, population, travel time) plus a learned **resource-type embedding**.

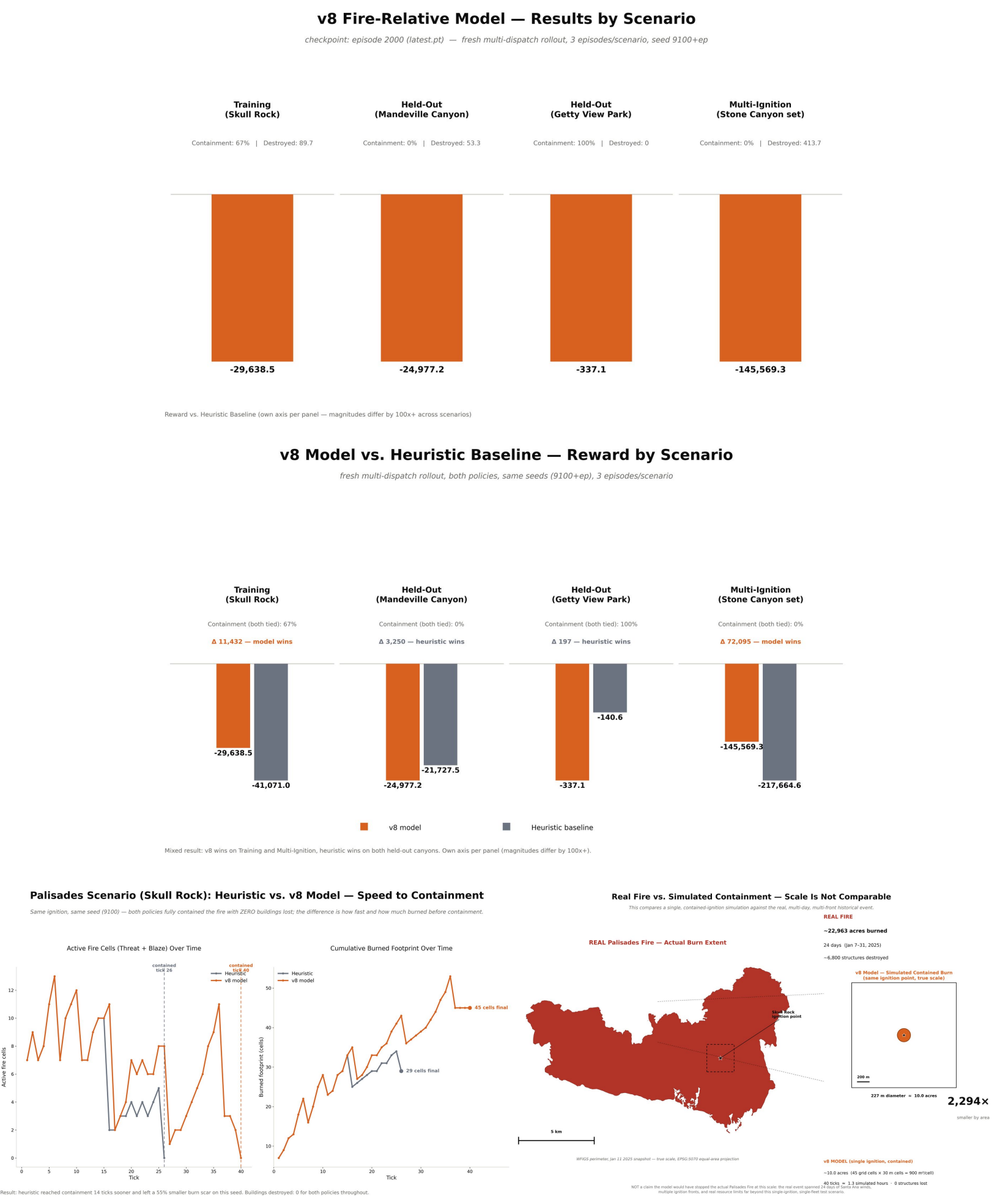
"Send the helicopter to the fire" means the same thing **no matter** where the fire started, so the **policy** transfers to new ignition points instead of **memorizing** one location.

This also **shrinks** the action space from **32** fixed zones down to about **6 relative options**, giving the model a stronger, denser training **signal per decision**.



Key Findings

- The **fire-relative model** (v8, checkpoint ep2000) was tested **against** the heuristic baseline on **four real scenarios** using full multi-dispatch, the **correct** way to evaluate this model.
- v8 **clearly outperforms** the heuristic on the two hardest, most dangerous scenarios: the training anchor (**Skull Rock**) and the **multi-ignition** scenario, cutting buildings destroyed substantially in both cases.
- On the two **smaller** held-out scenarios (**Mandeville Canyon**, **Getty View Park**), the heuristic is slightly ahead on reward, though containment is **identical** between the two policies on **every** scenario tested.
- On a **matched** tick-by-tick trace of the **real Palisades** scenario, both policies fully contained the fire with **zero buildings lost**. The heuristic did it slightly **faster** (26 ticks, 29-cell scar) than the model (40 ticks, 45-cell scar), showing room to **improve** on speed even where outcomes match.
- Overall: a real, honest, mixed result. v8 wins where it matters most (the **harder scenarios**), ties on **containment everywhere**, and still has ground to make up on **efficiency** in the easier cases.



Conclusion & Limitations

- Redesigning** the action space, not tuning the RL algorithm, was what **actually solved** the generalization problem. Absolute zone indices **memorized** one ignition point; **resolving** fire-relative semantic candidates let the same policy **transfer** to new locations.
- The **seven-version** investigation into the earlier collapse, though a series of "negative" results on their own terms, correctly diagnosed the **real bottleneck**: a GAE-driven critic instability in **near-terminal** states masked by a deeper structural issue in how actions were **represented**.
- The **final model (v8)** shows a real, honest, mixed result against the heuristic baseline: it **wins decisively** on the harder, more dangerous scenarios, **ties on containment everywhere**, and still **trails slightly in efficiency** on the easier cases.
- This is a **meaningful** step toward a wildfire dispatch agent that works on the **next fire**, not just the one it was **trained on**.

Future Work

- Finish **multi-dispatch training (v10)** to attack the fleet-capacity ceiling, and **swap** in real LANDFIRE fuel data for **more accurate** fire-spread physics.
- Validate** against real structure-damage data and **test** on more **ignition points** to get a **stronger** measure of generalization.

Sources & Acknowledgements

Data Sources

- OpenStreetMap contributors. OSMnx road network data.
- LA County GeoHub. LARIAC4 building footprint dataset.
- USGS. 3D Elevation Program (3DEP). 10m DEM.
- NOAA. ASOS station KSMO (Santa Monica Airport), historical weather data.
- U.S. Census Bureau. American Community Survey (ACS) block-group population data.
- LA County GeoHub / WFIGS. Palisades Fire perimeter data, 2025.
- LAFD. Fire Station Directory.

Literature

Tapley, A., Dotter, M., Doyle, M., Fennelly, A., Gandikota, D., Smith, S., Threet, M., & Welsh, T. (2023). Reinforcement Learning for Wildfire Mitigation in Simulated Disaster Environments. arXiv:2311.15925. Murray, L., Castillo, T., Martin de Diego, I., Weber, R., González-Olabarria, J. R., Garcia-Gonzalo, J., Weintraub, A., & Carrasco, J. (2025). Deep reinforcement learning for optimal firebreak placement in forest fire prevention. Applied Soft Computing. Mathur, S., et al. (2026). Spatiotemporal Wildfire Prediction and Reinforcement Learning for Helitack Suppression. arXiv:2601.14238.

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